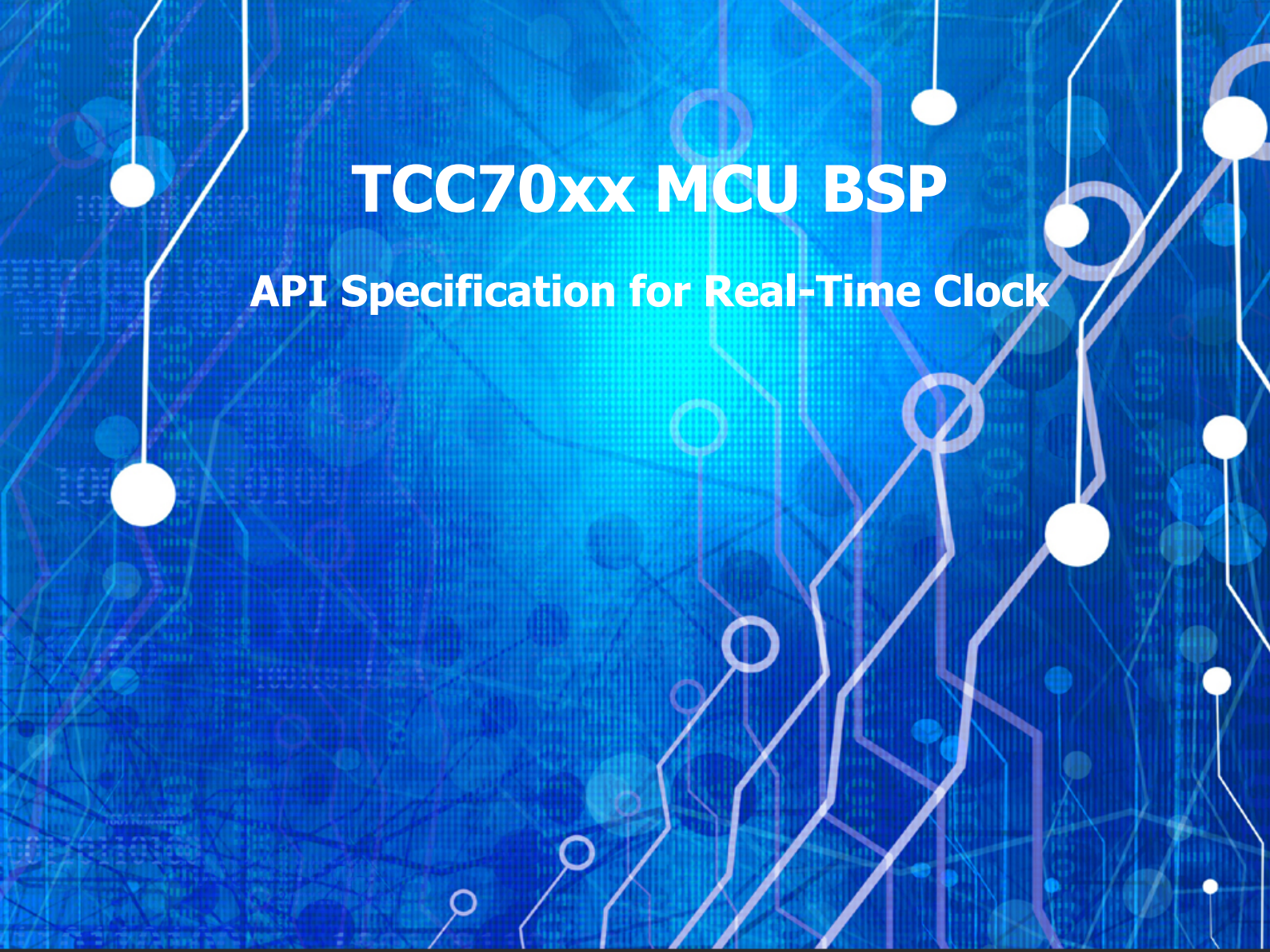




TCC70xx MCU BSP

API Specification for Real-Time Clock

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the Real-Time Clock device driver (hereinafter referred to as RTC driver) of TCC70xx MCU BSP. For more detailed information on RTC, refer to Chapter 27 Real-Time Clock (RTC) in "*TCC70xx Full Specification*"[1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

RTC	Real Time Clock
BCD	Binary Coded Decimal
PMWKUP	Power Management Wake Up

3 SOURCE FILES

3.1 Location of Source Files

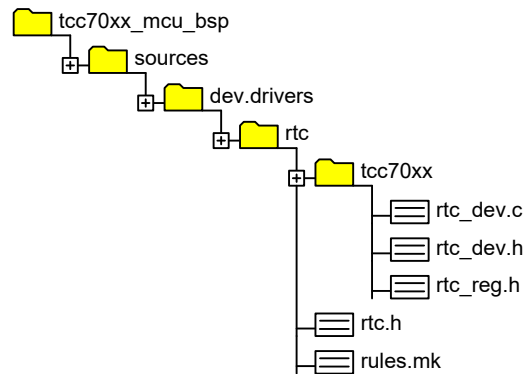


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `rtc_dev.c`
 - RTC driver source
- `rtc_dev.h`
 - RTC driver header
- `rtc_reg.h`
 - RTC register define header
- `rtc.h`
 - RTC header for parameter define and interface
- `rules.mk`
 - Rules used for build

4 ENUMERATIONS

4.1 RTCDay_t

Enumeration variables to manage a week. This value can be used as the 3rd parameter in `RTC_SetTime()` and `RTC_SetAlarm()`.

Declaration

```
typedef enum
{
    RTC_DAY_SUN    = 0,
    RTC_DAY_MON    = 1,
    RTC_DAY_TUE    = 2,
    RTC_DAY_WED    = 3,
    RTC_DAY_THU    = 4,
    RTC_DAY_FRI    = 5,
    RTC_DAY_SAT    = 6
} RTCDay_t;
```

Description

<i>Value</i>	<i>Description</i>
<code>RTC_DAY_[name]</code>	<i>Value that defines a week as a number</i>

Remark

None

4.2 RTCIrq_t

Enumeration variables to manage interrupt of RTC. This value can be used as parameter in RTC_EnableAlarm().

Declaration

```
typedef enum
{
    RTC_IRQ_DISABLE    = 0,
    RTC_IRQ_ALARM      = 1,
    RTC_IRQ_PMWKUP     = 2
} RTCIrq_t;
```

Description

Value	Description
RTC_IRQ_DISABLE	Disable IRQ
RTC_IRQ_ALARM	Enable ALARM IRQ
RTC_IRQ_PMWKUP	Enable PMWKUP IRQ

Remark

None

5 APIs

5.1 RTC_Init

This function is called from *bsp.c*, and RTC is initialized in BSP initialization.

Declaration

```
void RTC_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

5.2 RTC_SetTime

Sets time on RTC.

Declaration

```
void RTC_SetTime
(
    uint32 uiYear,
    uint32 uiMon,
    uint32 uiDay,
    uint32 uiDate,
    uint32 uiHour,
    uint32 uiMin,
    uint32 uiSec
);
```

Parameters

Value	Description
uiYear	[in] Decimal type year value Example: 2021UL
uiMon	[in] Decimal type month value Example: 12UL
uiDay	[in] Decimal type week day value. Refer to RTCDay_t . Example: RTC_DAY_MON
uiDate	[in] Decimal type date value Example: 30UL
uiHour	[in] Decimal type hour value Example: 12UL
uiMin	[in] Decimal type minute value Example: 59UL
uiSec	[in] Decimal type second value Example: 45UL

Return

None

Remark

None

5.3 RTC_GetTime

Gets time on RTC.

Declaration

```
void RTC_GetTime
(
    uint32 *   uiYear,
    uint32 *   uiMon,
    uint32 *   uiDay,
    uint32 *   uiDate,
    uint32 *   uiHour,
    uint32 *   uiMin,
    uint32 *   uiSec
);
```

Parameters

Value	Description
uiYear	[out] Return the decimal type year value Example: 2021UL
uiMon	[out] Return the decimal type month value Example: 12UL
uiDay	[out] Return the decimal type week day value. Refer to RTCDay_t . Example: RTC_DAY_MON
uiDate	[out] Return the decimal type date value Example: 30UL
uiHour	[out] Return the decimal type hour value Example: 12UL
uiMin	[out] Return the decimal type minute value Example: 59UL
uiSec	[out] Return the decimal type second value Example: 45UL

Return

None

Remark

None

5.4 RTC_ClearTime

Clears time on RTC.

Declaration

```
void RTC_ClearTime  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

5.5 RTC_SetAlarm

Sets an alarm on RTC.

Declaration

```
void RTC_SetAlarm
(
    uint32      uiYear,
    uint32      uiMon,
    uint32      uiDay,
    uint32      uiDate,
    uint32      uiHour,
    uint32      uiMin,
    uint32      uiSec,
    RTCHandler  fHandler
);
```

Parameters

Value	Description
uiYear	[in] Decimal type year value Example: 2021UL
uiMon	[in] Decimal type month value Example: 12UL
uiDay	[in] Decimal type week day value. Refer to RTCDay_t . Example: RTC_DAY_MON
uiDate	[in] Decimal type date value Example: 30UL
uiHour	[in] Decimal type hour value Example: 12UL
uiMin	[in] Decimal type minute value Example: 59UL
uiSec	[in] Decimal type second value Example: 45UL

Return

None

Remark

None

5.6 RTC_EnableAlarm

Enables an alarm on RTC.

Declaration

```
void RTC_EnableAlarm
(
    RTCIrq_t    tMode
);
```

Parameters

Value	Description
tMode	[in] Alarm type. Refer to RTCIrq_t. ■ RTC_IRQ_ALARM: Alarm interrupt (ALMINT) in normal operation mode ■ RTC_IRQ_PMWKUP: Wake-up (PMWKUP) signal generation: Supported in the power-down mode (PWDN)

Return

None

Remark

None

5.7 RTC_ClearAlarm

Clears an alarm on RTC.

Declaration

```
void RTC_ClearAlarm  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6 HOW TO USE RTC DRIVER

6.1 Initialize RTC Driver

```
{  
    RTC_Init();  
}
```

6.2 Set/Get RTC Time

```
{  
    uint32 uiYear;  
    uint32 uiMon;  
    uint32 uiDay;  
    uint32 uiDate;  
    uint32 uiHour;  
    uint32 uiMin;  
    uint32 uiSec;  
  
    RTC_GetTime(&uiYear, &uiMon, &uiDay, &uiDate, &uiHour, &uiMin, &uiSec);  
  
    uiYear = 2021UL;  
    uiMon = 1UL;  
    uiDay = (uint32)RTC_DAY_MON;  
    uiDate = 19UL;  
    uiHour = 12UL;  
    uiMin = 34UL;  
    uiSec = 56UL;  
  
    RTC_SetTime(uiYear, uiMon, uiDay, uiDate, uiHour, uiMin, uiSec);  
}
```

6.3 Set Alarm

```
static void RTC_TEST_AlarmHandler
(
    void
)
{
    RTC_ClearAlarm();
    RTC_EnableAlarm(RTC_IRQ_DISABLE);
}

...

{
    RTC_EnableAlarm(RTC_IRQ_DISABLE);

    RTC_SetAlarm
    (
        uiYear, uiMon, uiDay, uiDate, uiHour, uiMin, uiSec, (RTCHandler)RTC_TEST_AlarmHandler
    );

    RTC_EnableAlarm(RTC_IRQ_ALARM);
}
```

7 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

8 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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